

Robustness of Ordinary Least Squares and Ridge Estimators in the Presence of Outliers: A Simulation Study

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Abstract

Multicollinearity in linear regression often leads to unstable and unreliable parameter estimates. Ridge regression, a biased estimation method, is widely used to address this issue by producing more stable coefficient estimates. This study evaluates the top 16 ridge parameter estimators along with seven additional estimators introduced in subsequent research. These 23 methods were compared against Ordinary Least Squares (OLS) and Generalized Ridge (GR) regression across various multicollinearity levels using simulated data with and without outliers under diverse parametric conditions. Results (Mean Square Error) show that certain ridge estimators yield reliable performance in small samples with high correlations (≈ 0.95) when outliers are absent. However, in the presence of outliers, some estimators proved more effective due to sample sensitivity. GR demonstrated strong performance in large datasets, though their robustness diminished under extreme outlier conditions.

Keywords: MSE; Multicollinearity; OLS; Outliers; Ridge regression; Simulation

1. Introduction

Linear regression is a frequently utilized methodology for analysing data to construct a model depicting the relationship between the dependent variable and one or more independent variables (Herawati et al., 2024). During this type of analysis, it is common to encounter specific issues and hypotheses regarding the model. One important assumption in regression analysis is the absence of multicollinearity, as violating this assumption can render the model unreliable for estimating population parameters (Damodar N. Gujarati, 2013, Herawati et al., 2018). In multiple regression analysis, multicollinearity occurs when one independent variable is correlated with another independent variable. This can result in inaccurate estimates of the regression coefficients, increased standard errors, reduced partial t-tests, statistically insignificant p-values, and diminished predictability of the model (Gujarati, 2013).

Addressing high multicollinearity is crucial as it can lead to inaccurate decisions and an increased likelihood of accepting the wrong hypothesis. It is essential to find the most suitable method to deal with multicollinearity (Gibbons, 1981). One effective approach to overcome this issue is by using the shrinkage method to reduce the estimated coefficients, also known as regularization.

Ridge Regression stabilizes the regression coefficient in the presence of multicollinearity issues by introducing a level of bias to the regression estimate (Li et al., 2010). This reduces the standard error and provides a more precise estimation of the regression coefficient compared to the Ordinary Least Squares (OLS) method (Kibria & Lukman, 2020).

Hoerl and Kennard (1970) were the first to propose the concept of ridge regression to address the issue of multicollinearity in engineering data. They discovered that there exists a non-zero value of k (ridge parameter) for which the MSE of the ridge regression estimator is smaller than the variance of the OLS estimator (Hoerl & Kennard, 1970). Over the years, numerous researchers have delved into this area of study, devising and suggesting various estimators for k . To cite a few, Dempster et

al., 1977; Gibbons, 1981; Hoerl et al., 1975; Hoerl & Kennard, 1970; J. F. & P, 1976; Khalaf & Shukur, 2005; Kibria, 2003; Kibria et al., 2012a; Månsson et al., 2010; McDonald & Galarneau, 1975; McDonald & Schwing, 1973; Muniz & Kibria, 2009; Walker & Birch, 1988 and very recently Arashi & Valizadeh, 2015a, 2015b; Aslam, 2014a, 2014b; Ayinde & Lukman, 2016; A. V. Dorugade, 2014; Emmert-Streib & Dehmer, 2019; Kibria & Lukman, 2020; Hefnawy & Farag, 2014; Herawati et al., 2018, 2024; Kibria, 2022; Lukman et al., 2018, 2019; Lukman & Ayindez, 2017; Melkumova & Shatskikh, 2017; Mermi et al., 2024; Yüzbaşı et al., 2020 Hoque & Kibria, 2023; Hoque & Kibria, 2024, among others.

Mermi et al. (2024) compared 366 different ridge parameter estimators that were proposed at different times in terms of the number of independent variables (p), sample size (n), the correlation coefficient between independent variables (ρ), and standard deviation of errors (σ) (Mermi et al., 2024). Here, we have incorporated a total of 16 best estimators out of 366, with the exclusion of the robust estimator. Furthermore, we have applied seven estimators recommended by Kibria et al. (2003, 2023) tailored for asymmetric populations.

The objective of this paper is to provide a thorough analysis to compare 23 ridge parameter estimators with OLS and GR. Our analysis not only focused on MSE but also examined how the models behaved in the presence of multicollinearity and outliers. Our goal is to contribute to the existing literature on ridge regression by providing a comprehensive analysis of estimator performance, particularly in simulation scenarios where outliers have a significant impact on model estimates.

2. Statistical Methodology

2.1 Ordinary Least Squares Method

Let, Y is a column vector representing the dependent variable; X is a matrix of independent variables, and β is a parameter vector. The ε column vector represents the error terms. Then, the following is the multiple linear regression model,

$$Y = X\beta + \varepsilon$$

where Y is a column vector representing the dependent variable; X is a matrix of independent variables, and β is a parameter vector. The ε column vector represents the error terms. The ordinary least squares estimator of β is obtained as,

$$\hat{\beta}_{OLS} = (X^T X)^{-1} X^T Y$$

2.2 Ridge Regression Method

Hoerl and Kennard (1970) first introduced the ridge regression estimator as a solution to the multicollinearity issue. This approach minimizes the sum of squared residuals while applying a constraint on the total sum of the coefficient squares,

$$\hat{\beta}_{Ridge} = \underset{\beta}{\operatorname{argmin}} (Y - X\beta)^T (Y - X\beta) + k \| \beta \|^2$$

where $\| \beta \|^2 = \sum_{i=1}^p \beta_i^2$. Taking the above equation and finding its derivative with respect to β and setting the outcome to zero,

$$\hat{\beta}_{Ridge} = (X^T X + kI)^{-1} X^T Y, k \geq 0$$

where k is known as ridge or shrinkage estimator and need to be estimated from observed data.

2.3 Generalized Ridge (GR) Regression

Yang and Emura (2017) proposed GR regression by changing from uniform shrinkage to non-uniform shrinkage that replaced the identity matrix I_p with the diagonal matrix $\hat{M}(\Delta)$ (Yang & Emura, 2017). The following is the proposed GR method.

$$\hat{\beta}_{GR} = \{X^T X + k\hat{M}(\Delta)\}^{-1} X^T y$$

where $\Delta \geq 0$ is the threshold parameter and $k > 0$ is the shrinkage parameter. The diagonal components of $\hat{M}(\Delta) = \text{diag}\{\hat{m}_1(\Delta), \dots, \hat{m}_p(\Delta)\}$, where (Emura et al., 2024)

$$\hat{m}_i(\Delta) = \begin{cases} 1/2 & \text{if } z_i \geq \Delta \\ 1 & \text{if } z_i < \Delta \end{cases}$$

where $z_i = \hat{\beta}_i^0 / \text{SD}(\hat{\beta}^0)$, and $\text{SD}(\hat{\beta}^0) = \left\{ \sum_{i=1}^p (\hat{\beta}_i^0 - \sum_{i=1}^p \hat{\beta}_i^0 / p)^2 / (p-1) \right\}^{1/2}$ for $i = 1, \dots, p$, and $\hat{\beta}^0 = (\hat{\beta}_1^0, \dots, \hat{\beta}_p^0)^T$, defined as $\hat{\beta}_i^0 = X_i^T y / X_i^T X_i$, where X_i is the i -th column of X . And $\hat{\beta}^0$ is a compound covariate estimator (Chen & Emura, 2017).

2.4 Ridge parameters

In a recent study, Merimi et al. (2024) conducted a comparison of 366 different ridge parameter estimators that were proposed at different times. This study selected the top 16 estimators from Merimi et al. (2024), excluding the robust estimator. These selected estimators are k_1 through k_4 and k_{10} through k_{21} (Table 1). Additionally, the study added seven estimators, which are k_5 through k_9 proposed by Kibria (2023) and k_{22} and k_{23} by Kibria and Lukman (2020), specifically tailored for asymmetric populations. The 23 ridge parameters used in our study, including both simulation-based and empirical examples, are presented in Table 1.

Table 1: Existing best estimators based on the Monte Carlo Simulation

Estimators	Citation
$k_1 : \max\left(\frac{1}{q_i}\right)$	(Alkhamisi & Shukur, 2007)
$k_2 : \max\left(\frac{s^2}{\hat{\beta}_i^2}\right)$	(Alkhamisi et al., 2006)
$k_3 : \frac{\hat{\sigma}^2}{\min(\frac{\hat{\sigma}^2}{\hat{\alpha}_i^2} + \frac{1}{\lambda_i})}$	(Dorugade, 2014)
$k_4 : \sqrt{p \sum_{i=1}^p \left(\frac{\max(\lambda_i) \hat{\sigma}^2}{(n-p) \hat{\sigma}^2 + \max(\lambda_i) \hat{\alpha}_i^2} \right)}$	(Lukman & Ayindez, 2017)
$k_5 : \text{median} \left(\frac{\hat{\sigma}^2}{\hat{\beta}_i^2} + \frac{1}{\lambda_i} \right)$	(Kibria, 2023)
$k_6 : \max \left(\frac{\hat{\sigma}^2}{\hat{\beta}_i^2} + \frac{1}{\lambda_i} \right)$	(Kibria, 2023)
$k_7 : \min \left(\frac{\hat{\sigma}^2}{\hat{\beta}_i^2} + \frac{1}{\lambda_i} \right)$	(Kibria, 2023)
$k_8 : \text{geometric. mean} \left(\frac{\hat{\sigma}^2}{\hat{\beta}_i^2} + \frac{1}{\lambda_i} \right)$	(Kibria, 2023)

Estimators	Citation
k_9 : harmonic. mean $\left(\frac{\hat{\sigma}^2}{\hat{\beta}_i^2} + \frac{1}{\lambda_i} \right)$	(Kibria, 2023)
k_{10} : median $\left(\frac{\max(\lambda_i)\hat{\sigma}^2}{(n-p)\hat{\sigma}^2 + \lambda_i\hat{\alpha}_i^2} \right)$	(Lukman & Ayindez, 2017)
k_{11} : max $\left(\frac{\max(\lambda_i)\hat{\sigma}^2}{(n-p)\hat{\sigma}^2 + \max(\lambda_i)\hat{\alpha}_i^2} \right)$	(Lukman et al., 2018)
k_{12} : $\sqrt{p \sum_{i=1}^p \left(\frac{\lambda_i\hat{\sigma}^2}{(n-p)\hat{\sigma}^2 + \lambda_i\hat{\alpha}_i^2} \right)}$	(Lukman & Ayindez, 2017)
k_{13} : $\frac{1}{p} \sum_{i=1}^p \frac{\max(\lambda_i)\hat{\sigma}^2}{(n-p)\hat{\sigma}^2 + \lambda_i\hat{\alpha}_i^2}$	(Alkhamisi et al., 2006)
k_{14} : max $\left(\frac{1}{\frac{\lambda_i\hat{\sigma}^2}{(n-p)\hat{\sigma}^2 + \lambda_i\hat{\alpha}_i^2}} \right)$	(Muniz et al., 2012)
k_{15} : $\left(\prod_{i=1}^p \frac{\max(\lambda_i)\hat{\sigma}^2}{(n-p)\hat{\sigma}^2 + \lambda_i\hat{\alpha}_i^2} \right)^{\frac{1}{p}}$	(Muniz et al., 2012)
k_{16} : max $\left(1/\sqrt{\frac{\lambda_i\hat{\sigma}^2}{(n-p)\hat{\sigma}^2 + [\max(\lambda_i)]\hat{\alpha}_i^2}} \right)$	(Lukman et al., 2018)
k_{17} : max $\left(1/\frac{\lambda_i\hat{\sigma}^2}{(n-p)\hat{\sigma}^2 + \lambda_i\max(\hat{\alpha}_i^2)} \right)$	(Lukman et al., 2018)
k_{18} : $\prod_{i=0}^p \left(\frac{1}{q_i} \right)^{\frac{1}{p}}$	(Kibria et al., 2012)
k_{19} : max $\left(1/\frac{\lambda_i\hat{\sigma}^2}{(n-p)\hat{\sigma}^2 + \lambda_i(\max(\hat{\alpha}_i))^2} \right)$	(Lukman et al., 2018)
k_{20} : $\frac{1}{\sqrt{\frac{2\hat{\sigma}^2}{\max(\lambda_i)\max(\hat{\alpha}_i^2)}}}$	(Ayinde & Lukman, 2016)
k_{21} : $\frac{1}{\sqrt{\frac{2p\hat{\sigma}^2}{\max(\lambda_i)\sum_{i=1}^p \hat{\alpha}_i^2}}}$	(Ayinde & Lukman, 2016)
k_{22} : $\hat{\sigma}p^{\left(1+\frac{p}{n}\right)}$	(Kibria & Lukman, 2020)
k_{23} : $\hat{\sigma} \times \max \left(p^{\left(1+\frac{p}{n}\right)}, p^{\left(1+\frac{1}{p}\right)} \right)$	(Kibria & Lukman, 2020)

3. Simulation study

As the theoretical comparison is not feasible, we conducted a Monte Carlo simulation study using the R program in this section.

3.1 Simulation Technique

The data generation process for the models was carried out according to a widely used method as described below (Gibbons, 1981; Kibria, 2003, Nayem et al., 2024).

$$x_{li} = (1 - \rho^2)^{1/2} u_{li} + \rho u_{l,p+1}, l = 1, 2, \dots, n; i = 1, 2, \dots, p$$

where ρ denotes the correlation coefficient between any two explanatory variables, u_{jli} denotes the independent pseudo-random variable obtained from standard normal distribution and p denotes the number of explanatory variables. Moreover, the dependent variable Y was obtained through the following equation (Alheety et al., 2025).

$$Y_l = \beta_0 + \beta_1 X_{l1} + \beta_2 X_{l2} + \dots + \beta_p X_{lp} + \varepsilon_l, l = 1, 2, \dots, n$$

where, ε_l is the error term, i.e. $\varepsilon_l \sim N(0, \sigma^2)$.

For modelling, two different values of explanatory variables, $p = 5$ and 10 , two different values of the correlation coefficients between the explanatory variables, $\rho = 0.9$ and 0.95 , four different values of the sample sizes, $n = 20, 30, 50$, and 100 , and the error variance $\sigma^2 = 1$ were taken. The data generation process of the explanatory variables was carried out with the help of the values determined for p , ρ , n , and σ . The experiment was repeated, $N = 5000$ times. In the response variable, 10 % and 20% outliers were added.

The MSE was calculated by comparing the estimated coefficients of each model with the true coefficients (assumed $\beta = I_p$) and averaged MSE was calculated based on the following formula over all simulations.

$$AMSE(\hat{\beta}_i^*) = \frac{1}{N} \sum_{i=1}^P (\hat{\beta}_i - \beta)' (\hat{\beta}_i - \beta)$$

where $\hat{\beta}_i^*$ is any of the OLE, Ridge, Lasso, or EN estimators, β is the true parameter (assumed $\beta = I_p$), and p is the number of predictors.

3.2 Results and Discussion

3.2.1 Without Outlier

Table 2 reveals that, for a correlation coefficient of 0.90 and no outliers, the MSE of the estimators behaves differently based on sample size and number of predictors. In smaller samples (e.g., 20 or 30 observations) with five predictors, estimators such as k_3 , k_{10} , k_{11} , k_{13} , k_{15} , and GR tend to show relatively low MSE. For small samples (e.g., 20 or 30 observations) the estimators like k_4 , k_{12} , and k_{21} are more effective when the number of predictors is unchanged. In contrast, for larger samples, regardless of the variance level, the GR is generally preferred due to their efficiency, particularly in the presence of a correlation of 0.90. It's important to note that OLS does not perform well when collinearity is present among the independent variables.

Table 2: MSE values of the estimators with correlation = 0.9

Estimator	P = 5, $\sigma^2 = 1$				P = 10, $\sigma^2 = 1$			
	20	30	50	100	20	30	50	100
OLS	0.684	0.637	0.616	0.606	0.726	0.701	0.639	0.611
k_1	0.798	0.851	0.910	0.960	0.806	0.807	0.843	0.917
k_2	0.806	0.841	0.884	0.935	0.853	0.852	0.868	0.915
k_3	0.574	0.536	0.520	0.514	0.701	0.590	0.537	0.514
k_4	0.662	0.616	0.569	0.532	0.806	0.764	0.704	0.615
k_5	0.704	0.670	0.640	0.619	0.793	0.768	0.729	0.653
k_6	0.640	0.617	0.600	0.591	0.692	0.653	0.627	0.594
k_7	0.813	0.768	0.710	0.662	0.925	0.885	0.833	0.764
k_8	0.712	0.678	0.645	0.621	0.805	0.771	0.726	0.661
k_9	0.688	0.663	0.637	0.618	0.761	0.736	0.702	0.648
k_{10}	0.573	0.535	0.515	0.506	0.663	0.587	0.537	0.511
k_{11}	0.573	0.535	0.515	0.506	0.662	0.587	0.537	0.511
k_{12}	0.633	0.590	0.552	0.524	0.752	0.707	0.651	0.578

Estimator	P = 5, $\sigma^2 = 1$				P = 10, $\sigma^2 = 1$			
	20	30	50	100	20	30	50	100
k ₁₃	0.573	0.535	0.515	0.506	0.662	0.587	0.537	0.511
k ₁₄	0.961	0.972	0.978	0.986	0.969	0.980	0.989	0.993
k ₁₅	0.573	0.535	0.515	0.506	0.662	0.587	0.537	0.511
k ₁₆	0.821	0.830	0.836	0.856	0.881	0.871	0.876	0.885
k ₁₇	0.962	0.973	0.979	0.986	0.969	0.980	0.989	0.993
k ₁₈	0.785	0.846	0.909	0.960	0.761	0.788	0.837	0.916
k ₁₉	0.962	0.973	0.979	0.986	0.969	0.980	0.989	0.993
k ₂₀	0.648	0.611	0.578	0.554	0.760	0.718	0.666	0.599
k ₂₁	0.628	0.595	0.567	0.547	0.724	0.685	0.640	0.585
k ₂₂	0.785	0.779	0.775	0.772	0.904	0.879	0.855	0.827
k ₂₃	0.785	0.773	0.755	0.739	0.904	0.879	0.855	0.827
GR	0.575	0.536	0.516	0.507	0.668	0.594	0.546	0.517

Table 3 shows that the performance of OLS deteriorates significantly as the correlation coefficient rises from 0.90 to 0.95. Consistent with the findings in Table 2, for small samples and five predictors, estimators such as k₁₀, k₁₁, k₁₃, k₁₅, and GR exhibit relatively low MSE. While an increase in the number of predictors leads to higher MSE, these estimators maintain stable performance for small sample sizes. For larger samples, regardless of variance, the GR remain the preferred choices due to their operational efficiency.

Table 3: MSE values of the estimators with correlation = 0.95

Estimator	P = 5				P = 10			
	20	30	50	100	20	30	50	100
OLS	0.731	0.653	0.618	0.606	0.963	0.792	0.671	0.616
k ₁	0.808	0.845	0.908	0.959	0.863	0.844	0.847	0.911
k ₂	0.816	0.836	0.884	0.938	0.884	0.874	0.870	0.908
k ₃	0.610	0.547	0.522	0.515	0.905	0.669	0.563	0.517
k ₄	0.690	0.631	0.577	0.536	0.855	0.813	0.742	0.636
k ₅	0.741	0.687	0.649	0.626	0.860	0.838	0.789	0.674
k ₆	0.663	0.628	0.607	0.597	0.765	0.711	0.667	0.609
k ₇	0.867	0.808	0.736	0.677	0.951	0.933	0.891	0.822
k ₈	0.752	0.700	0.657	0.630	0.867	0.840	0.781	0.689
k ₉	0.715	0.677	0.647	0.626	0.823	0.798	0.745	0.668
k ₁₀	0.598	0.546	0.517	0.506	0.750	0.646	0.563	0.515
k ₁₁	0.597	0.546	0.517	0.506	0.749	0.645	0.563	0.515
k ₁₂	0.660	0.604	0.558	0.527	0.808	0.762	0.694	0.597
k ₁₃	0.598	0.546	0.517	0.506	0.749	0.645	0.563	0.515
k ₁₄	0.975	0.979	0.982	0.987	0.977	0.987	0.993	0.996
k ₁₅	0.598	0.546	0.517	0.506	0.749	0.645	0.563	0.515
k ₁₆	0.853	0.849	0.850	0.864	0.915	0.903	0.900	0.906
k ₁₇	0.975	0.980	0.982	0.987	0.977	0.987	0.993	0.996
k ₁₈	0.794	0.839	0.906	0.959	0.816	0.823	0.839	0.910
k ₁₉	0.975	0.980	0.982	0.987	0.977	0.987	0.993	0.996
k ₂₀	0.682	0.629	0.587	0.560	0.833	0.787	0.717	0.623
k ₂₁	0.661	0.612	0.575	0.552	0.801	0.753	0.690	0.606
k ₂₂	0.798	0.780	0.778	0.777	0.919	0.893	0.861	0.829
k ₂₃	0.798	0.775	0.759	0.745	0.919	0.893	0.861	0.829
GR	0.601	0.548	0.519	0.508	0.761	0.658	0.581	0.526

3.2.2 With outlier

Table 4 shows that the MSE of the estimators is reported for correlated predictors at 0.90 and a 10% outlier present in the data. The addition of outliers is seen to result in an elevated MSE in comparison to the values outlined in Table 2. Notably, in the case of small sample sizes, it is evident that k₂₀, k₂₂,

and k_{23} demonstrate low MSE. However, as the number of predictors increase, k_{22} and k_{23} perform well. Furthermore, for small predictors, the estimators k_{16} , k_{22} , and k_{23} demonstrate better performance. For a high number of predictors, k_{16} , k_{22} , and k_{23} are recommended. Additionally, in the case of large samples, the GR is recommended due to their ease of operation.

Table 4: MSE values of the estimators with correlation = 0.90 and 10% outlier

Estimator	$P = 5, \sigma^2 = 1$				$P = 10, \sigma^2 = 1$			
	20	30	50	100	20	30	50	100
OLS	1.635	1.048	0.856	0.706	3.548	1.804	1.030	0.887
k_1	0.829	0.831	0.878	0.935	0.893	0.862	0.847	0.896
k_2	0.852	0.817	0.829	0.877	0.924	0.879	0.857	0.877
k_3	1.496	0.996	0.736	0.596	3.231	1.700	0.998	0.676
k_4	0.928	0.784	0.678	0.588	0.966	0.845	0.745	0.639
k_5	0.878	0.762	0.687	0.617	1.011	0.847	0.765	0.674
k_6	1.072	0.820	0.684	0.598	1.832	1.116	0.788	0.635
k_7	0.871	0.840	0.801	0.712	0.927	0.914	0.901	0.856
k_8	0.867	0.766	0.699	0.627	0.993	0.845	0.769	0.686
k_9	0.918	0.768	0.682	0.613	1.179	0.882	0.749	0.657
k_{10}	1.419	1.000	0.748	0.605	1.864	1.357	0.956	0.680
k_{11}	1.407	0.999	0.748	0.605	1.808	1.351	0.956	0.680
k_{12}	0.995	0.811	0.686	0.589	1.106	0.887	0.745	0.630
k_{13}	1.430	1.002	0.748	0.605	1.921	1.376	0.958	0.680
k_{14}	0.946	0.955	0.965	0.976	0.963	0.975	0.984	0.990
k_{15}	1.438	1.003	0.748	0.605	1.996	1.384	0.959	0.680
k_{16}	0.829	0.799	0.792	0.795	0.904	0.878	0.858	0.854
k_{17}	0.948	0.956	0.966	0.976	0.965	0.975	0.984	0.990
k_{18}	0.812	0.814	0.871	0.934	0.911	0.836	0.826	0.891
k_{19}	0.948	0.956	0.966	0.976	0.965	0.975	0.984	0.990
k_{20}	0.814	0.734	0.660	0.587	0.897	0.829	0.744	0.646
k_{21}	0.858	0.756	0.664	0.584	0.953	0.846	0.736	0.631
k_{22}	0.816	0.765	0.738	0.709	0.905	0.875	0.837	0.793
k_{23}	0.816	0.762	0.724	0.681	0.905	0.875	0.837	0.793
GR	0.891	0.797	0.699	0.598	1.000	0.897	0.772	0.640

Table 5 illustrates the MSE of the estimators when there is a 0.95 correlation among predictors and a 10% outlier in the data. It is evident that the introduction of outliers leads to a higher MSE compared to the findings in Table 3. In scenarios with the number of predictors is five, it is observed that k_{18} , k_{20} , k_{22} , and k_{23} yield low MSE, but k_{18} is noted to be an inconsistent estimator. However, as the number of predictors increases, k_{22} and k_{23} exhibit strong performance. In cases of small samples, k_{20} , k_{22} , and k_{23} demonstrate robust performance when the correlation coefficient is approximately 0.95. For large samples, the GR is recommended due to their user-friendly operation.

Table 5: MSE values of the estimators when correlation = 0.95 and 10% outlier

Estimator	$P = 5, \sigma^2 = 1$				$P = 10, \sigma^2 = 1$			
	20	30	50	100	20	30	50	100
OLS	2.335	1.204	0.806	0.710	6.649	2.966	1.467	0.871
k_1	0.852	0.840	0.873	0.935	0.932	0.903	0.864	0.894
k_2	0.873	0.826	0.823	0.879	0.971	0.913	0.870	0.879
k_3	2.077	1.125	0.778	0.601	5.939	2.711	1.385	0.753
k_4	1.009	0.806	0.687	0.591	1.108	0.919	0.795	0.665
k_5	0.920	0.780	0.691	0.624	1.111	0.903	0.816	0.706
k_6	1.246	0.857	0.693	0.602	2.756	1.442	0.902	0.663

Estimator	P = 5, $\sigma^2 = 1$				P = 10, $\sigma^2 = 1$			
	20	30	50	100	20	30	50	100
k ₇	0.888	0.858	0.809	0.728	0.947	0.934	0.920	0.890
k ₈	0.901	0.780	0.701	0.636	1.088	0.906	0.814	0.717
k ₉	0.991	0.787	0.684	0.620	1.459	0.986	0.802	0.682
k ₁₀	1.795	1.110	0.792	0.609	2.420	1.711	1.224	0.754
k ₁₁	1.776	1.108	0.792	0.609	2.388	1.709	1.223	0.753
k ₁₂	1.112	0.841	0.698	0.592	1.386	1.002	0.809	0.656
k ₁₃	1.823	1.114	0.793	0.609	2.524	1.762	1.233	0.754
k ₁₄	0.965	0.969	0.971	0.978	0.974	0.985	0.991	0.994
k ₁₅	1.848	1.115	0.793	0.609	2.720	1.792	1.235	0.754
k ₁₆	0.860	0.827	0.802	0.804	0.936	0.917	0.889	0.881
k ₁₇	0.967	0.970	0.971	0.978	0.975	0.985	0.991	0.994
k ₁₈	0.837	0.821	0.865	0.934	0.979	0.889	0.840	0.887
k ₁₉	0.967	0.970	0.971	0.978	0.975	0.985	0.991	0.994
k ₂₀	0.848	0.752	0.665	0.592	0.950	0.888	0.795	0.676
k ₂₁	0.897	0.772	0.670	0.588	1.014	0.906	0.789	0.658
k ₂₂	0.844	0.779	0.737	0.714	0.933	0.903	0.851	0.801
k ₂₃	0.844	0.777	0.723	0.687	0.933	0.903	0.851	0.801
GR	0.968	0.825	0.713	0.602	1.129	0.985	0.849	0.674

Table 6 presents the MSE of the estimators when there is a 0.90 correlation among predictors and a 25% outlier in the data. A comparison with Table 4 indicates that as the proportion of outliers in the data set increases, the MSE also increases. Specifically, when the number of predictors (p) is five for small sample sizes, estimators k₇, k₁₆, k₂₂, and k₂₃ display small MSE. However, some estimators, such as k₁, k₁₄, k₁₈, and k₁₉ show inconsistency under similar conditions. Additionally, it is observed that the MSE of k₂₂ and k₂₃ increase as the number of predictors increase. The performance of k₇, k₁₆, k₂₂, and k₂₃ are found to be better for small sample sizes. Furthermore, for a high number of predictors, these estimators are recommended. However, for large sample sizes, the GR is suggested due to their ease of use.

From Table 7, the MSE of the estimators is presented for predictors with a correlation of 0.95 and a 25% outlier in the data. A comparison to Table 5 reveals that as the outlier in the dataset increased, the MSE also increased. Among the estimators, k₇, k₂₀, k₂₂, and k₂₃ demonstrate low MSE for the lower number of predictors in small samples. However, some estimators, such as k₁, k₁₄, k₁₆, k₁₇, and k₁₉, exhibit low MSE for small samples but lack consistency. Notably, for small predictors k₇, k₂₂, and k₂₃ exhibit consistent and better performance. Furthermore, for a high number of predictors, k₇, k₁₆, k₂₂, and k₂₃ are recommended. It is observed that the GR method result in lower MSE when dealing with outliers in large samples.

Table 6: MSE values of the estimators when correlation = 0.90 and 25% outlier

Estimator	P = 5, $\sigma^2 = 1$				P = 10, $\sigma^2 = 1$			
	20	30	50	100	20	30	50	100
OLS	3.243	2.104	1.281	0.913	7.119	3.515	1.818	1.057
k ₁	0.910	0.877	0.865	0.915	0.953	0.931	0.880	0.886
k ₂	1.046	0.992	0.842	0.834	1.082	0.983	0.896	0.861
k ₃	2.985	1.984	1.232	0.798	6.547	3.315	1.753	1.040
k ₄	1.675	1.389	1.029	0.754	1.447	1.170	0.970	0.806
k ₅	1.424	1.193	0.904	0.710	1.588	1.148	0.920	0.772
k ₆	2.148	1.542	1.033	0.733	3.913	2.224	1.297	0.874

Estimator	P = 5, $\sigma^2 = 1$				P = 10, $\sigma^2 = 1$			
	20	30	50	100	20	30	50	100
k ₇	1.020	0.948	0.866	0.791	0.985	0.954	0.916	0.889
k ₈	1.371	1.139	0.879	0.710	1.522	1.142	0.919	0.771
k ₉	1.633	1.282	0.926	0.709	2.158	1.412	1.009	0.787
k ₁₀	2.875	2.011	1.264	0.811	3.896	2.705	1.686	1.045
k ₁₁	2.849	2.005	1.263	0.811	3.692	2.679	1.684	1.045
k ₁₂	1.893	1.500	1.071	0.764	1.920	1.418	1.073	0.845
k ₁₃	2.898	2.016	1.264	0.811	4.016	2.756	1.692	1.045
k ₁₄	0.942	0.945	0.952	0.966	0.963	0.971	0.979	0.986
k ₁₅	2.923	2.018	1.264	0.811	4.279	2.787	1.693	1.045
k ₁₆	0.947	0.891	0.804	0.769	0.960	0.932	0.878	0.841
k ₁₇	0.945	0.948	0.953	0.967	0.965	0.972	0.979	0.986
k ₁₈	0.958	0.877	0.854	0.913	1.112	0.979	0.868	0.876
k ₁₉	0.945	0.948	0.953	0.967	0.965	0.972	0.979	0.986
k ₂₀	1.149	1.054	0.866	0.698	1.071	1.005	0.897	0.766
k ₂₁	1.315	1.180	0.932	0.719	1.249	1.124	0.964	0.800
k ₂₂	1.040	0.944	0.799	0.711	0.963	0.935	0.869	0.795
k ₂₃	1.040	0.951	0.804	0.700	0.963	0.935	0.869	0.795
GR	1.324	1.191	0.943	0.745	1.392	1.176	1.027	0.838

Table 7: MSE values of the estimators when correlation = 0.95 and 25% outlier

Estimator	P = 5, $\sigma^2 = 1$				P = 10, $\sigma^2 = 1$			
	20	30	50	100	20	30	50	100
OLS	4.718	2.750	1.461	0.945	13.045	6.190	2.821	1.337
k ₁	0.943	0.900	0.871	0.915	0.985	0.966	0.911	0.890
k ₂	1.058	1.009	0.881	0.897	1.146	1.032	0.932	0.872
k ₃	4.275	2.554	1.403	0.826	11.875	5.723	2.651	1.298
k ₄	1.965	1.581	1.116	0.773	1.857	1.347	1.049	0.864
k ₅	1.588	1.304	0.957	0.718	1.803	1.208	0.957	0.813
k ₆	2.707	1.847	1.133	0.749	6.338	3.263	1.616	0.986
k ₇	1.037	0.977	0.882	0.794	1.018	0.975	0.932	0.898
k ₈	1.515	1.240	0.920	0.719	1.743	1.228	0.969	0.809
k ₉	1.924	1.449	0.991	0.720	2.931	1.704	1.114	0.843
k ₁₀	3.832	2.559	1.435	0.843	5.469	3.820	2.360	1.303
k ₁₁	3.793	2.548	1.434	0.843	5.335	3.796	2.355	1.302
k ₁₂	2.296	1.750	1.175	0.785	2.731	1.745	1.196	0.927
k ₁₃	3.892	2.571	1.436	0.843	5.632	3.965	2.385	1.303
k ₁₄	0.961	0.962	0.960	0.970	0.978	0.981	0.989	0.992
k ₁₅	3.970	2.577	1.436	0.843	6.288	4.085	2.393	1.303
k ₁₆	0.979	0.917	0.824	0.778	0.990	0.963	0.913	0.872
k ₁₇	0.963	0.964	0.962	0.970	0.979	0.982	0.989	0.992
k ₁₈	1.026	0.907	0.859	0.912	1.195	1.047	0.907	0.878
k ₁₉	0.963	0.964	0.962	0.970	0.979	0.982	0.989	0.992
k ₂₀	1.214	1.103	0.901	0.707	1.123	1.058	0.942	0.805
k ₂₁	1.406	1.250	0.980	0.730	1.321	1.175	1.008	0.847
k ₂₂	1.146	1.003	0.826	0.717	1.032	0.991	0.906	0.815
k ₂₃	1.146	1.013	0.835	0.706	1.032	0.991	0.906	0.815
GR	1.531	1.351	1.018	0.762	1.699	1.426	1.156	0.947

4. Conclusion

This study conducted a comprehensive evaluation of multiple linear regression models in the presence of multicollinearity. We employed ridge regression as a biased estimation technique to obtain more precise estimates of the regression coefficients. From the 366 proposed estimators for the ridge parameter, we focused on the top 16, along with seven estimators suggested by other researchers, and compared their performance against traditional OLS and modern regularization techniques such as GR regression. To simulate real-world conditions, we introduced outliers at 10% and 25% levels to assess their impact on estimator performance. The primary comparison criterion was the MSE, a standard metric for evaluating estimator accuracy.

Our findings revealed nuanced insights into the selection of ridge parameter estimators under various parametric conditions. Specifically, with small sample sizes and a high degree of multicollinearity (correlation close to 0.95) in the absence of outliers, the estimators k_{10} , k_{11} , k_{13} , and k_{15} proved to be reliable, balancing bias and variance to produce lower MSEs. However, in the presence of outliers, particularly with small sample sizes and high variance, the estimators k_7 , k_{16} , k_{22} , and k_{23} performed better, making them the preferred choices in such situations.

For larger sample sizes, the performance dynamics shifted, the GR regression emerging as robust options. These methods consistently delivered lower MSEs across different levels of multicollinearity, except in cases where significant outliers combined with large variances. Under such challenging conditions, even these robust methods saw performance declines, indicating the need for careful estimator selection in the presence of extreme outliers.

Acknowledgments

The authors wish to acknowledge the contributions of previous researchers whose foundational work has significantly informed and guided this study.

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